Matthew (Moslem) Shokrolahi

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SUMMARY

Applied Machine Learning Scientist with a Ph.D. in Computer Engineering and 5+ years of experience developing and evaluating machine learning solutions across banking, e-commerce, and computer vision. Strong foundations in deep learning, predictive modeling, NLP, and statistical experimentation, with measurable improvements in recommendation relevance, detection precision, and operational efficiency. Experienced in Python, SQL, AWS SageMaker, Databricks, and GPU-accelerated computing, with skills in API development, CI/CD, and ML evaluation pipelines. Collaborates with scientists, engineers, and business stakeholders to translate complex analytical problems into practical AI solutions.

CORE EXPERTISE

Applied Data Science | Predictive Modeling | Deep Learning | NLP & Text Analytics | Statistical Experimentation | Model Development & Evaluation | ML Software Engineering | Cloud & High-Performance Computing | MLOps & CI/CD | Explainable & Responsible AI | Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Applied Machine Learning Scientist: Model Risk Management & Compliance Oversight | TD Bank | Toronto, ON **| Jan 2026 – Present**

- Research, benchmark, and validate machine learning and deep learning models across credit, fraud, insurance, and marketing applications, assessing predictive performance, stability, explainability, and fairness.
- Develop scalable ML evaluation pipelines using statistical hypothesis testing, confidence intervals, and comparative benchmarking to assess model reliability and inform improvement recommendations.
- Use AWS SageMaker and Databricks to evaluate enterprise ML models, investigate performance issues, and communicate findings that support technical and business decisions.
- Collaborate with model developers, data scientists, and engineering teams to investigate model limitations and recommend improvements to performance, robustness, and lifecycle controls.
- Research LLM, vision-language model, and agentic AI evaluation methods, translating emerging techniques into practical frameworks for assessing model behavior and reliability.

AI Consultant | KnowQuest | Toronto, ON **| Sep 2023 - Sep 2025**

- Developed a CLIP-based multimodal classification solution for detecting AI-generated and mismatched customer review images, improving fake-review detection precision by 18% and supporting more reliable e-commerce analytics.
- Designed and developed an LLM-powered retrieval-augmented generation framework for product recommendations and customer feedback analysis, improving mean reciprocal rank by 23% and increasing recommendation relevance.

Research Assistant | UAI Lab, Queen's University | Kingston, ON **| May 2021 - Jul 2026**

- Developed interpretable and controllable diffusion models to improve demographic fairness, generation safety, and image consistency through deep learning research and experimental evaluation.
- Investigated U-Net and transformer-based foundation models, including SDXL, PixArt-α, FLUX.1, and SD3.5, using attention-based analysis to understand model behavior and guide controllable generation.
- Developed continual learning and machine unlearning algorithms using deep metric learning, conducting large-scale training, benchmarking, and experimental evaluation on GPU-accelerated HPC infrastructure.

Machine Vision Engineer | SAESPADAN | Isfahan **| Sep 2018 - May 2021**

- Designed and implemented a real-time industrial computer vision system using an ensemble of ResNet and EfficientNet models, reducing human classification errors by 40% and increasing operational throughput by approximately 30% through optimized model inference and weighted ensemble predictions.

TECHNICAL SKILLS

Programming & Data: Python, SQL, C++, NumPy, Pandas

Machine Learning & Statistics: Supervised Learning, Deep Learning, Predictive Modeling, Statistical Hypothesis Testing, Confidence Intervals, Experimental Evaluation, Model Benchmarking, Explainability, Fairness & Bias Evaluation

NLP & Generative AI: Large Language Models, Text Analytics, Retrieval-Augmented Generation, Vision-Language Models, Hugging Face Transformers, LangChain, LangGraph, Diffusion Models

Frameworks: PyTorch, TensorFlow, scikit-learn, OpenCV

ML Engineering & APIs: FastAPI, REST APIs, Docker, Git/GitHub, GitHub Actions, CI/CD, MLflow, LangFuse

Cloud & High-Performance Computing: AWS, Databricks, Linux, GPU-Accelerated Training, SLURM

Data Visualization: Matplotlib, Seaborn, Plotly, Model Performance Visualization

EDUCATION

Ph.D. in Computer Engineering | **Queen's University, Kingston, ON** | May 2021 - Jul 2026

Dissertation: Adaptive and Responsible Machine Learning: From Incremental Learning to Unlearning and Safe Generation

M.Sc. in Computer Engineering | **Isfahan University of Technology (IUT), Isfahan** | Sep 2015

INDEPENDENT PROJECT

Smart Resume Screening: Developing an NLP-based workflow to ingest PDF/DOCX resumes and technical job descriptions, extract structured skills, experience, education, and certifications, and separate identity information from ranking features.

Agentic AI Evaluation 1.0 Bootcamp | Vector Institute | Feb–Apr 2026

- Designed and implemented a fraud analytics AI agent using Google ADK, Gemini 2.5 Pro, SQL, and Python sandbox execution to automate transaction analysis and support fraud investigation.
- Developed a three-level evaluation framework covering prediction correctness (Precision, Recall, F1), tool-use compliance and data leakage prevention, and LLM-as-a-Judge assessment of reasoning quality and evidence grounding.
- Integrated LangFuse for agent tracing and evaluation, identified ground-truth leakage risks, and implemented safeguards that eliminated observed leakage cases in offline testing.

SELECTED PUBLICATIONS

- Shokrolahi, Sayedmoslem, Jae-Mo Kang, and Il-Min Kim. “Responsible Text-to-Image Diffusion: Interpretable and Linearly Controllable Semantics for Fair and Safe Generation.” Forty-third International Conference on Machine Learning **(ICML), 2026.**
- Shokrolahi, Sayedmoslem, Jae-Mo Kang, and Il-Min Kim. “Learning Where It Matters: Responsible and Interpretable Text-to-Image Generation with Background Consistency.” Transactions on Machine Learning Research **(TMLR), 2026.**
- Shokrolahi, Sayedmoslem, and Il-Min Kim. “MaxDiv: Zero-Shot Machine Unlearning via Distributionally Divergent Erasing Samples.” IEEE Transactions on Artificial Intelligence **(TAI), 2025**.
- Shokrolahi, Sayedmoslem, and Il-Min Kim. “Combating Inter-Task Confusion and Catastrophic Forgetting by Metric Learning and Re-Using a Past Trained Model.” Transactions on Machine Learning Research **(TMLR), 2025.**